

KARRATHA, Australia

WMO: 953070

Lat: 20.7097S

Lon: 116.7742E

Elev: 6

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	12.8	-3.3	2.9	20.8	-1.1	3.5	21.2	11.8	20.4	11.0	22.3	3.1	170	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.5	39.4	23.1	38.1	23.4	36.8	23.6	28.5	32.6	27.9	32.2	27.4	31.9	6.5	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.6	23.6	30.8	26.8	22.5	30.4	26.2	21.7	30.2	92.5	32.3	89.5	32.2	87.2	32.1	36.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.6	11.0	10.0	DB	9.4	43.0	1.2	1.4	8.6	44.0	7.9	44.8	7.3	45.6	6.4	46.6
			WB	5.3	29.6	1.0	1.7	4.5	30.8	3.9	31.8	3.3	32.8	2.6	34.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	30.9	30.9	30.8	28.5	24.0	20.7	20.0	21.2	23.6	27.3	28.8	30.3	(6)
	DBStd	4.62	2.10	2.08	2.12	1.97	2.22	2.28	2.02	1.84	2.06	2.70	2.40	2.31	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	13	0	0	0	0	0	5	7	2	0	0	0	0	(9)
	CDD10.0	5978	649	584	644	553	434	321	311	346	409	536	563	628	(10)
	CDD18.3	2949	391	351	386	303	176	76	59	89	159	277	313	370	(11)
	CDH23.3	35918	5404	4903	5442	3795	1529	441	382	734	1495	3152	3764	4878	(12)
(13)	CDH26.7	17951	2982	2714	3077	1884	513	62	48	146	543	1531	1871	2580	(13)
(14)	Wind	WSAvg	5.4	6.4	5.8	5.3	4.4	4.4	4.5	4.4	4.6	5.3	6.0	6.7	6.6 (14)
(15) Precipitation	PrecAvg	251	26	62	46	20	27	34	14	5	1	0	1	9	(15)
	PrecMax	506	125	219	288	125	150	202	86	34	9	12	8	80	(16)
	PrecMin	43	0	0	0	0	0	0	0	0	0	0	0	0	(17)
	PrecStd	119	36	60	65	32	38	50	21	9	3	2	2	19	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	41.0	40.3	38.1	34.0	30.2	29.9	31.4	35.0	39.1	40.1	40.5	(19)
		MCWB	23.2	23.9	23.1	22.0	19.8	19.6	17.5	17.8	18.9	20.4	20.9	22.0	(20)
	2%	DB	38.4	38.1	38.4	36.2	32.2	28.4	27.9	29.5	32.9	36.9	37.6	38.2	(21)
		MCWB	24.8	25.2	23.9	21.7	19.9	18.5	16.9	17.3	18.4	20.3	21.7	23.3	(22)
	5%	DB	36.5	36.5	36.9	34.8	30.8	27.1	26.7	28.3	31.4	35.1	35.9	36.5	(23)
		MCWB	25.6	25.3	24.1	21.8	19.8	18.0	16.7	17.0	18.4	20.7	22.0	23.7	(24)
	10%	DB	34.9	35.0	35.5	33.6	29.5	25.9	25.5	27.1	29.8	33.4	34.2	34.9	(25)
		MCWB	26.1	25.8	24.4	21.9	19.7	17.7	16.5	16.7	18.1	20.7	22.0	24.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	29.1	28.9	27.5	25.7	24.4	22.5	21.7	23.0	26.1	26.8	28.0	(27)
		MCDB	33.3	33.4	32.9	31.1	27.9	25.7	24.0	25.7	27.6	30.3	31.4	32.7	(28)
	2%	WB	28.6	28.2	28.0	26.3	24.4	22.4	20.7	20.8	21.9	24.5	25.6	27.2	(29)
		MCDB	32.6	32.5	32.3	30.5	27.2	24.7	23.9	25.2	27.3	29.3	30.7	32.1	(30)
	5%	WB	27.9	27.7	27.4	25.6	23.3	21.3	19.6	19.8	21.0	23.6	24.9	26.6	(31)
		MCDB	32.1	32.1	31.8	30.0	26.9	24.1	23.6	24.6	26.7	29.4	30.5	31.6	(32)
	10%	WB	27.3	27.2	26.9	24.9	22.2	20.2	18.5	18.9	20.3	22.8	24.2	26.0	(33)
		MCDB	31.9	31.9	31.5	29.7	26.5	23.7	23.0	24.1	26.2	29.1	30.1	31.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.5	8.6	10.1	10.3	10.0	11.2	12.4	12.3	11.4	10.0	8.4	(35)
		MCDBR	10.0	10.0	11.0	12.0	12.0	11.1	12.6	13.7	14.3	13.4	12.5	10.8	(36)
	5% WB	MCWBR	4.1	4.3	4.7	5.5	5.7	5.6	6.2	6.5	6.4	5.6	5.1	4.5	(37)
		MCWBR	7.7	8.1	8.4	9.2	9.1	7.8	9.9	11.7	12.5	11.0	9.8	8.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.402	0.397	0.378	0.353	0.335	0.322	0.317	0.314	0.333	0.357	0.359	0.378	(40)
		taud	2.506	2.533	2.575	2.600	2.586	2.609	2.596	2.565	2.512	2.472	2.505	2.524	(41)
	Ebn at Noon		944	936	929	914	894	888	904	940	958	963	978	967	(42)
		Edh at Noon	115	110	102	92	86	81	84	94	106	116	114	113	(43)
(44) All-Sky Solar Radiation	RadAvg		7.46	7.14	6.38	5.56	4.65	4.21	4.57	5.65	6.68	7.55	8.04	8.03	(44)
	RadStd		0.45	0.49	0.46	0.30	0.35	0.30	0.29	0.16	0.25	0.17	0.19	0.32	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.46	N/A	N/A	N/A	+0.48	+0.56	N/A	N/A	N/A	+158

Nomenclature: See separate page